Assignment 2-3

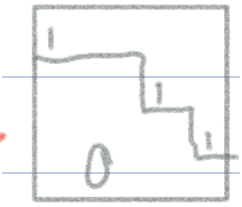
① Gauss-Jordan Reduction Calculation A2-1. A2-4

② Parameter in Augmented Matrix ⇒ Consistent & inconsistent

③ Matrix multiplication and Algebrical properties

④ Consistency theorem for Linear system

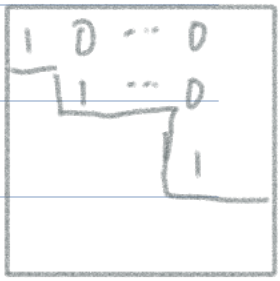
The process of using elementary row operations to transform a matrix into reduced row echelon form is called ***Gauss-Jordan reduction 高斯若尔当消元法***.



The reduced row echelon form of a matrix is **unique**.

Gaussian Elimination → r.e.f / r.r.e.f

Gauss-Jordan reduction → r.r.e.f



1. Indicate whether each of the following systems is consistent, if it is consistent, use Gauss-Jordan reduction to solve the system:

(a)

$$\begin{aligned} 3x_1 + 2x_2 - x_3 &= 4 \\ x_1 - 2x_2 + 2x_3 &= 1 \\ 11x_1 + 2x_2 + x_3 &= 11 \end{aligned}$$

(b)

$$x_1 + 3x_2 + x_3 + x_4 = 3$$

4. Given the linear systems

(i) $\begin{aligned} x_1 + 2x_2 &= 2 \\ 3x_1 + 7x_2 &= 8 \end{aligned}$ (ii) $\begin{aligned} x_1 + 2x_2 &= 1 \\ 3x_1 + 7x_2 &= 7 \end{aligned}$

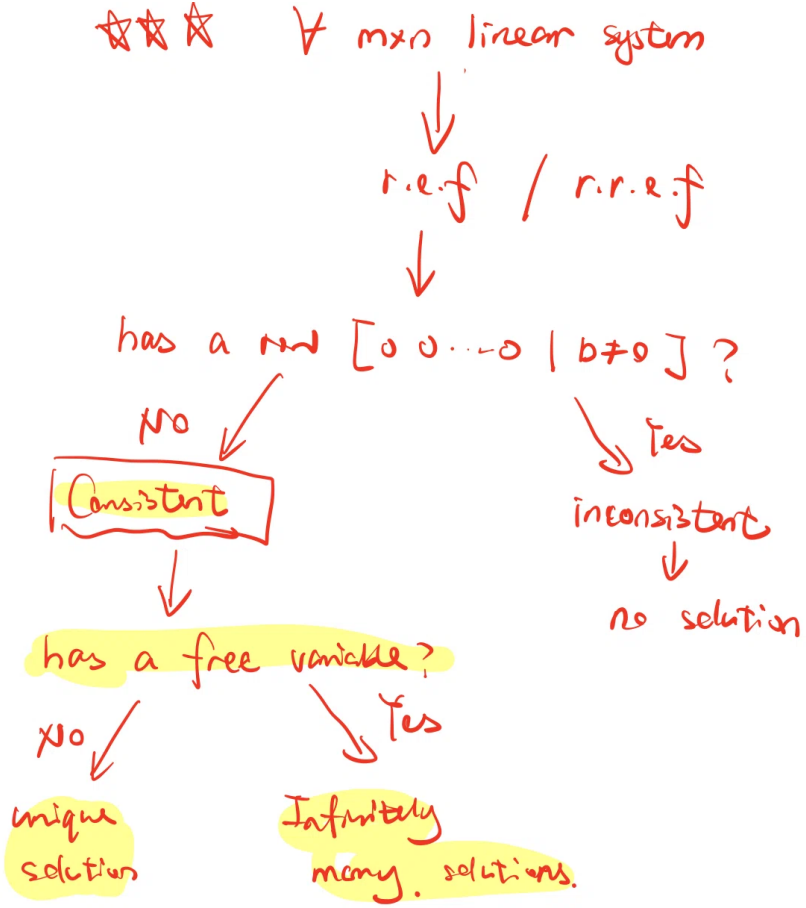
[A | b1] [A | b2]

solve both systems by incorporationg the right-hand sides into a 2×2 matrix B and computing the reduced row echelon form of

parellel

$$(A|B) = \left[\begin{array}{cc|cc} 1 & 2 & 2 & 1 \\ 3 & 7 & 8 & 7 \end{array}\right]$$

[A | b] [A | b1 b2] [A | B] B=[b1 b2]



2. Find the values or ranges of k and h such that the following system has

$$x_1 + hx_2 = 2$$
$$4x_1 + 8x_2 = k$$

(a) no solution;
(b) the unique solution;
(c) infinitely many solutions.

3. Consider a linear system whose augmented matrix is of the form

$$\left(\begin{array}{ccc|c} 1 & 1 & 3 & 2 \\ 1 & 2 & 4 & 3 \\ 1 & 3 & a & b \end{array}\right)$$

(a) For what values of a and b will the system have infinitely many solutions?
(b) For what values of a and b will the system be inconsistent?

Matrix Multiplication

Definition

If the number of columns in a matrix A equals the number of rows in a matrix B, then product matrix AB exists.

Let A: $m \times n$ matrix, B: $n \times k$ matrix,

The **product matrix** $C = AB$ is a $m \times k$ matrix, with entries

$$c_{ij} = \begin{bmatrix} a_{i1} & a_{i2} & \cdots & a_{in} \end{bmatrix} \begin{bmatrix} b_{1j} \\ b_{2j} \\ \vdots \\ b_{nj} \end{bmatrix} = a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{in}b_{nj}$$

Theorem: Properties of Transpose

Let *A* and *B* be matrices and *c* be a scalar. Assume that the sizes of the matrices are such that the operations can be performed.

1. $(A + B)^T = A^T + B^T$ — *Transpose of a sum*
2. $(cA)^T = cA^T$ — *Transpose of a scalar multiple*
3. $(AB)^T = B^T A^T$ — *Transpose of a product*
4. $(A^T)^T = A$

1. For each of the pairs of matrices that follow, determine whether it is possible to multiply the first matrix times the second. If it is possible, perform the multiplication.

(a) $\begin{bmatrix} 3 & 5 & 1 \\ -2 & 0 & 2 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 1 & 3 \\ 4 & 1 \end{bmatrix}$

(b) $\begin{bmatrix} 4 & 2 \\ 6 & -4 \\ 8 & -6 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \end{bmatrix}$

(c) $\begin{bmatrix} 1 & 4 & 3 \\ 0 & 1 & 4 \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} 3 & 2 \\ 1 & 1 \\ 4 & 5 \end{bmatrix}$

$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 2 & 3 \end{bmatrix} \begin{bmatrix} 2 & -1 & 4 \\ 1 & 0 & -2 \\ 0 & 3 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 5 \\ 2 & 9 & -1 \end{bmatrix}$

5. Let matrices A, B and C be

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 2 & 3 \end{bmatrix}, \quad B = \begin{bmatrix} 2 & -1 & 4 \\ 1 & 0 & -2 \\ 0 & 3 & 1 \end{bmatrix}, \quad C = \begin{bmatrix} 0 & 2 \\ -1 & 0 \\ 3 & 1 \end{bmatrix}$$

Compute AB, B^2, AC, CA and $B^T A^T$.

$B^T A^T = (AB)^T$

$= \begin{bmatrix} 2 & 3 \\ 2 & 9 \\ 5 & 1 \end{bmatrix}$

6. Let

$$A = \begin{bmatrix} 1 & 0 & -1 & 2 \\ 0 & 3 & 1 & -1 \\ 2 & 4 & 0 & 3 \\ -3 & 1 & -1 & 2 \end{bmatrix}, B = \begin{bmatrix} 1 & 2 \\ 3 & -1 \\ 0 & -2 \\ 4 & 1 \end{bmatrix} \text{ and } C = \begin{bmatrix} 3 & -2 & 0 & 5 \\ 1 & 0 & -3 & 4 \end{bmatrix}.$$

(a) Does the matrix $D = ABC$ exist? If so, what is the $(3,4)$-entry d_{34} of D?

(b) Does the matrix $E = BAC$ exist? If so, what is the $(2,2)$-entry e_{22} of E?

☆☆☆ Linear combination of vectors.

If $a_1, a_2 \cdots a_n \in \mathbb{R}^m$ and $c_1, c_2, \cdots c_n \in \mathbb{R}$ (scalars)

then $c_1a_1 + c_2a_2 + \cdots + c_na_n$

is a linear combination of vectors $a_1 \cdots a_n$.

scalar multiplication, addition

☆☆☆ Consistency theorem for linear systems ($\forall$ mxn)

(a) A linear system $Ax=b$ is consistent if and only if $[a_1\ a_2 \cdots a_n]x = b$

$$\downarrow \uparrow \quad [a_1\ a_2 \cdots a_n]\begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}$$

(b) b can be written as a linear combination of column vectors of A mxn

$x_1a_1 + x_2a_2 + \cdots + x_na_n = b$

$a_1, a_2, \cdots, a_n$

$\exists c_1, c_2, \cdots, c_n \in \mathbb{R}$, such that $b = c_1a_1 + c_2a_2 + \cdots + c_na_n$

2. Let $A = \begin{bmatrix} 1 & 2 & s \\ 3 & -1 & 2 \end{bmatrix}$ and $\mathbf{x} = \begin{bmatrix} t \\ s \\ 1 \end{bmatrix}$. If $A\mathbf{x} = \begin{bmatrix} 6 \\ 8 \end{bmatrix}$, find s and t.

3. Let A be a 3×4 matrix and let

$$\mathbf{b} = 5\mathbf{a}_1 + \mathbf{a}_2 - 3\mathbf{a}_4$$

$b = 5a_1 + a_2 + 0\cdot a_3 - 3a_4$

where $\mathbf{a}_i$ is the i-th column of matrix A. Will the system $A\mathbf{x} = \mathbf{b}$ be consistent? If it is consistent, how many solutions will the system have?

$Ax=b$ consistent.

$x = \begin{bmatrix} 5 \\ 1 \\ 0 \\ -3 \end{bmatrix}$

$[5\ \ 1\ \ 0\ \ -3]$

4. Let A be a 3×3 matrix and suppose that

$$\mathbf{a}_1 - 4\mathbf{a}_2 - 7\mathbf{a}_3 = \mathbf{0}$$

where $\mathbf{a}_i$ is the i-th column of matrix A and $\mathbf{0}$ is the zero vector. Will the system $A\mathbf{x} = \mathbf{0}$ be inconsistent? If it is consistent, how many solutions will the system have?